WORKED SOLUTION

STEP I 2006, Question 1

Pure • Difference of Two Squares

> **STEP I 2006, Question 1**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/06-s1-q1

The key idea

If $(n+m)^2 - 33\,127 = k^2$, then $33\,127 = (n+m)^2 - k^2$ is a difference of two squares, and so it factorises as $(n+m-k)(n+m+k)$. This is Fermat's old trick for factorising: start just above $\sqrt{N}$ and step upwards until $a^2 - N$ is a square. Running the argument backwards, every factorisation of $33\,127$ gives a value of m, so counting the values of m comes down to counting factorisations.

Solution

Finding n

Since $180^2 = 32\,400$, the square root of $33\,127$ is a little above 180. We find

$$182^2 = 33\,124 < 33\,127 < 33\,489 = 183^2,$$

so $n = 182$. Note that $33\,127 = 182^2 + 3$, so it is not itself a perfect square.

A small value of m

We try $m = 1, 2, \ldots$ in turn:

$$183^2 - 33\,127 = 33\,489 - 33\,127 = 362, \quad \text{not a square } (19^2 = 361),$$
$$184^2 - 33\,127 = 33\,856 - 33\,127 = 729 = 27^2.$$

So $m = 2$ works, with $(n+m)^2 - 33\,127 = 27^2$.

Factorising $33\,127$

Rearranging and using the difference of two squares,

$$33\,127 = 184^2 - 27^2 = (184 - 27)(184 + 27) = 157 \times 211,$$

so we may take $p = 157$ and $q = 211$, both greater than 1.

Exactly two values of m

Both factors are prime. For 157: since $13^2 = 169 > 157$, we only need to test the primes $2, 3, 5, 7, 11$, and none divides 157. For 211: since $15^2 = 225 > 211$, we test $2, 3, 5, 7, 11, 13$, and none divides 211. So $33\,127 = 157 \times 211$ is the prime factorisation, and the only ways to write 33 127 as a product uv of positive integers with $u \le v$ are

$$1 \times 33\,127 \qquad \text{and} \qquad 157 \times 211.$$

Now suppose $(n+m)^2 - 33\,127 = k^2$ with $k \ge 0$. We take $a = n + m > 0$, which is the situation the question has in mind (we start at n and move upwards). Then

$$(a-k)(a+k) = 33\,127.$$

Here $a + k > 0$, so $a - k > 0$ as well, and $a - k \le a + k$. So $(a-k,\, a+k)$ must be one of the two pairs (u, v) above, and then

$$a = \frac{u+v}{2}, \qquad k = \frac{v-u}{2}.$$

These are integers because u and v are both odd.

> **Ask yourself** Why does each factorisation give exactly one m, and why can no other m sneak in?

The map runs both ways. Given a and k, the pair $(a-k, a+k)$ is a factorisation; given a factorisation (u, v), the formulas above recover a and k uniquely. So the values of a (and hence $m = a - 182$) correspond one-to-one with the factor pairs, and there are exactly two factor pairs.

- $(u, v) = (157, 211)$ gives $a = 184$, $k = 27$, so $m = 184 - 182 = 2$, the value we found.
- $(u, v) = (1, 33\,127)$ gives $a = 16\,564$, $k = 16\,563$, so

$$m = 16\,564 - 182 = 16\,382.$$

Hence there are exactly two values of m, namely $m = 2$ and $m = 16\,382$. As a check of the second, $16\,564^2 - 16\,563^2 = 16\,564 + 16\,563 = 33\,127$.

(If we also allowed $n + m$ to be negative, then a could be -184 or $-16\,564$, giving the mirror-image values $m = -366$ and $m = -16\,746$. The question clearly intends $n + m$ to be positive.)

Checks

A direct computer search over every m from 0 up to 16 564 finds that $(182+m)^2 - 33\,127$ is a perfect square only for $m = 2$ and $m = 16\,382$, and that the divisors of 33 127 are exactly $1, 157, 211, 33\,127$. By hand, $157 \times 211 = 157 \times 200 + 157 \times 11 = 31\,400 + 1\,727 = 33\,127$.

Common slips

- Forgetting the trivial factorisation $1 \times 33\,127$, and so finding only one value of m.
- Claiming that 157 and 211 are prime without checking: the argument that there are *exactly* two values depends on it.
- Not checking that u and v have the same parity. Here both are odd, so a and k are integers, but for an even number that is not a multiple of 4 no factorisation would work.

Beyond STEP

This is Fermat's factorisation method, which works quickly when N has two factors close to $\sqrt{N}$, as here. The same idea, of finding $a^2 \equiv k^2 \pmod{N}$ rather than $a^2 - N = k^2$ exactly, is the basis of the quadratic sieve, one of the fastest general-purpose factoring algorithms. The link between factorisations and representations as a difference of squares is a nice first taste of the elementary number theory in Davenport, *The Higher Arithmetic*.

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